

Sepehr Behtaji

ML Engineer | Computer Vision & Applied AI (Neurodiagnostic + FinTech)

 Sepehr.Behtaji@gmail.com

 07535658759

 UK

 linkedin

 github

 Google Scholar

Profile

---

Machine Learning / AI Engineer focused on computer vision and applied AI, with experience delivering end-to-end ML systems from data preparation and modelling through evaluation, explainability, and deployment. Strong Python background (feature engineering, supervised learning, validation, and model interpretation). Combines hands-on ML engineering with technical product thinking—translating user and business needs into reliable, measurable AI features.

Education

---

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023/01 – 2024/05<br>Middlesbrough, UK | <b>MSc in Bioinformatics; Data science applied to biology (Distinction)</b><br><i>Teesside University</i><br><i>Thesis: Predictive modelling of high-dimensional biological datasets</i> <ul style="list-style-type: none"><li>Integrated multiple large-scale sequencing datasets to analyse relationships between regulatory features and system-level output metrics</li><li>Built, evaluated and interpreted machine-learning models (XGBoost, Random Forest) in Python to identify key predictive features</li><li>Applied model interpretability techniques (SHAP) to translate complex models into clear, explainable insights.</li></ul> |
| 2019/09 – 2020/12<br>Nancy, France     | <b>MSc in RNA and Enzyme Sciences (Merit)</b><br><i>University of Lorraine</i><br>Oxytocin-related lncRNAs and target genes in breast cancer (expression profiling) <ul style="list-style-type: none"><li>Bioinformatics: gene selection from public datasets + pathway enrichment (KEGG/Enrichr)</li><li>Wet lab: RT-qPCR workflow (RNA extraction, cDNA synthesis, primer design, melt-curve/gel QC)</li></ul>                                                                                                                                                                                                                                 |
| 2014/07 – 2018/07<br>Tonekabon, Iran   | <b>BSc in Cellular and Molecular Biology, Genetics (First Class)</b><br><i>Azad University of Tonekabon</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

Experience

---

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025/05 – present<br>London, United Kingdom | <b>AI / ML Engineer &amp; Technical Product Manager</b><br><i>GoodFolio LTD</i> <ul style="list-style-type: none"><li>Developed and internally validated high-accuracy computer-vision models for GoodMind’s CT/MRI neurodiagnostic platform across stroke, dementia and brain tumour use cases.</li><li>Built CNN, 3D medical-imaging and classifier workflows for disease classification, abnormality localisation and diagnostic-support outputs.</li><li>Trained and evaluated dementia MRI models across multiple datasets, using confusion matrices, cross-dataset testing and Grad-CAM heatmaps to assess performance and explain model decisions.</li><li>Developed CT-stroke classification workflows to distinguish normal scans, ischemic stroke and haemorrhagic/bleeding cases, supporting urgent neuroimaging triage use cases.</li><li>Created ML-ready imaging cohorts from DICOM scans, clinical labels, metadata, timestamps and outcome proxies to support model training, evaluation and validation.</li><li>Translated clinical needs, dataset requirements and AI capabilities into product features, validation plans, roadmap priorities and pilot-ready workflows.</li><li>Developed and deployed AI agents for automated client outreach using Google ADK, Gemini, Vertex AI, Model Garden, Cloud Run and Vertex AI Agent Engine.</li><li>Built and evaluated ML components for rule-to-check generation and layout/vision-aware checks, producing explainable, auditable inspection outputs for regulated fintech/regtech firms.</li><li>Collaborated across AI, product, security and operations teams, while also supporting ISO 27001 audit-readiness, risk tracking and compliance documentation.</li></ul> |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024/01 – 2024/06<br>Darlington, UK | <b>MSc Researcher – Data Modelling &amp; Analytics (MSc dissertation project)</b><br><i>Teesside University</i> <ul style="list-style-type: none"> <li>• Built and evaluated supervised ML models on large, complex datasets to identify predictive and explainable patterns in gene-level translation outcomes across two treatment conditions.</li> <li>• Led an independent analytical project integrating multiple high-dimensional datasets to study relationships between regulatory features and system-level outputs</li> <li>• Designed and executed data processing pipelines for large sequencing-derived datasets, focusing on data quality, alignment and aggregation</li> <li>• Built and compared interpretable machine-learning models (Random Forest, XGBoost, AdaBoost) to identify key predictive features</li> <li>• Applied SHAP-based interpretability to translate complex models into clear, explainable insights</li> </ul> |
| 2023/03 – 2025/04<br>London, UK     | <b>Contractor (part-time) – Data &amp; Machine Learning Engineer</b><br><i>ATLASI Ltd (Client delivery / services)</i> <ul style="list-style-type: none"> <li>• Built and maintained ETL pipelines for high-volume datasets supporting client analytics, reporting, and model evaluation.</li> <li>• Used SQL and Python for ad-hoc analysis, data validation, and exception reporting.</li> <li>• Implemented data quality checks to reduce defects, improve cross-system consistency, and strengthen traceability of key fields.</li> <li>• Supported release/testing cycles by verifying data integrity across environments and preventing reporting/pipeline regressions.</li> <li>• Wrote and maintained end-to-end Gherkin test packs to validate workflows and pipeline behaviour before release.</li> <li>• Worked in cloud environments to run scalable, repeatable processing and scheduled validation tasks.</li> </ul>                   |
| 2021/10 – 2022/12<br>Tehran, Iran   | <b>Data Scientist / Bioinformatics Researcher</b><br><i>Geniran Research Laboratory</i> <ul style="list-style-type: none"> <li>• Built machine-learning models on large, complex datasets to identify predictive and explanatory patterns</li> <li>• Curated and integrated structured and unstructured features, developing experience in feature selection and model interpretation</li> <li>• Collaborated with multidisciplinary teams to translate analytical findings into testable hypotheses and insights</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2020/04 – 2020/12<br>Nancy, France  | <b>Research Intern – Data Analysis &amp; Validation</b><br><i>University of Lorraine</i> <ul style="list-style-type: none"> <li>• Combined computational analysis with experimental validation, strengthening skills in data quality control and interpretation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Professional Certificates

|                                                                                 |                                                                              |
|---------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Python programming</b><br>Sematec IT Training Center                         | <b>SQL Server 2022 Database Implementation</b><br>Sematec IT Training Center |
| <b>Google Agent Development Kit (ADK)-Build &amp; Deploy AI Agents</b><br>Udemy |                                                                              |

## Publications

|            |                                                                                                                                                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026/05/10 | <b>High-Accuracy Lung Region Segmentation and Injury Classification Using Deep Networks</b><br><i>Scientific Reports</i>                                                                                                                |
| 2026/04/20 | <b>The adaptive engagement framework: enhancing banking customer experience through AI-powered invisible marketing</b> <br><i>Scientific Reports</i> |

## Core Skills

---

### Machine Learning & AI

- Supervised learning: classification, regression, model evaluation and validation
- Algorithms: XGBoost, Random Forest, AdaBoost, SVM, KNN, Gradient Boosting, Logistic Regression
- Feature engineering, feature selection, dimensionality reduction and correlation analysis
- Model interpretability and explainability: SHAP, feature importance, bias assessment
- Cross-validation and performance reporting: ROC-AUC, F1-score, precision, recall, Dice coefficient
- Class imbalance handling and sampling strategies, including Random OverSampler

### Medical Imaging & Computer Vision

- CT/MRI medical-imaging analysis for stroke, dementia and brain tumour use cases
- Explainable medical-imaging AI using Grad-CAM heatmaps and confusion-matrix analysis
- CT/MRI classification workflows for dementia staging, stroke subtype detection and tumour-support use cases
- 3D medical-imaging workflows, CNN-based classification and diagnostic-support modelling
- Image segmentation and multi-class classification using U-Net and CNN backbones
- CNN architectures: ResNet50, VGG16, Xception
- Hybrid modelling: CNN feature extraction with classical ML classifiers
- Imaging-derived feature analysis, abnormality localisation and model benchmarking

### AI Product, Validation & Stakeholder Communication

- Translating clinical, technical and business requirements into AI product features
- Technical product management, roadmap support and pilot-ready workflow design
- Documentation of data structures, model assumptions, rules and processes
- Communicating complex analytical results to technical and non-technical stakeholders
- Statistical reasoning, experimental design and technical reporting

### Programming & Tools

- Python and R for data processing, automation and machine-learning workflows
- Python libraries: pandas, NumPy, scikit-learn, PyTorch
- Data visualisation and reporting: Matplotlib, Seaborn
- Version control: Git
- Collaboration and delivery tools: Jira
- Advanced Excel for data validation, manipulation and reporting

### ML Data Infrastructure & Cloud

- Data extraction, cleansing, preprocessing and validation for large real-world datasets
- Building and optimising data pipelines for high-dimensional and multimodal datasets
- Handling structured, unstructured and imaging-derived data
- SQL database querying, reconciliation and troubleshooting
- Cloud and scalable processing environments: Google Cloud, AWS, Azure, HPC
- Workflow automation and reproducibility using Nextflow and Jupyter Notebook

### AI Agents & Product Automation

- AI agent development, deployment and workflow orchestration
- Google ADK, Vertex AI Agent Engine, Gemini on Vertex AI, Model Garden and Cloud Run
- Agentic outreach automation and growth workflow support
- Production-oriented AI workflow design and cloud deployment